

Experience

- **Capra Robotics** March – Sept. 2026
R&D Robotics Engineer (Intern) Aarhus, Denmark
Enabled autonomous food-cart docking on the company's core market (18/18 attempts) by designing a cart pose estimation algorithm reaching sub-centimeter accuracy from 3D LiDAR data, under 45 ms per cycle on CPU.
Shipped to production a fully on-camera GDPR anonymization pipeline (OAK-D), cutting stream latency by 86% and porting YOLOX-S to an FP16 blob (PyTorch → ONNX → OpenVINO).
Added semantic understanding of traversable ground to the robot by training and deploying a segmentation model on Jetson Orin Nano (IoU 0.981) and fusing it into the Nav2 costmap, on a 1,840-image dataset built with 10× less manual labelling.
- **Safran Data Systems** Sept 2023 – March 2026
R&D Innovation Engineer (Apprentice) Les Ulis, France
Let flight-test engineers program embedded acquisition hardware in plain English with no C/C++ skills, by automating the full path from intent to cross-compiled ARM binary (Python/C++), co-authored a publication at ETTC 2025 (European Test and Telemetry Conference).
Engineered FPGA logic for real-time current and voltage acquisition on flight test equipment.
- **Blue Frog Robotics** May – July 2023
Engineer in Embedded Systems (Intern) Paris, France
Brought up an 8x8 ToF matrix sensor (I2C, C) on the Buddy robot, from driver-level integration to a gesture-detection algorithm triggering robot behaviours.

Education

- **INSA Lyon** 2023 – 2026
Master of Science in Electrical Engineering (Diplôme d'Ingénieur) Lyon, France
Specializing in Signal & Image Processing (TDSI).
Built a biosurveillance vision pipeline for BioMAÉ (YOLOv11-N, MobileNetV3, 90% accuracy) and co-authored a technical paper on it.
- **University of Toulon** 2021 – 2023
Bachelor of Technology (BUT GEII) Toulon, France
Specialized in Electronics and Embedded Systems (ESE); won the school robotics competition with an autonomous PIC32 robot written in C.

Projects & Competitions

- **Mini-Factory Hackathon (Hugging Face LeRobot)** June 2026
Winner, Most Creative Project Award Copenhagen, Denmark
Trained an ACT imitation learning policy (LeRobot, SO-101 arm) fusing vision with tactile data from an experimental piezoresistive sensor.
Built "LePotager", an end-to-end pipeline grasping fruits, probing their firmness and sorting them by ripeness, from data collection to a live demo in 48 hours.
- **RoboCup (MSL) & French Robotics Cup (Eurobot)** 2022 – 2023
Scientific Challenge 1st Place & MSL 3rd Place France
Kept two autonomous competition robots running through multi-day tournaments, owning mechanical maintenance and systemic reliability, and designed their chassis in Fusion 360.

Skills and Interests

Robotics: ROS 2, Zenoh, Nav2, NVIDIA Jetson, OpenVINO / ONNX, FPGA

Machine Learning: PyTorch, imitation learning (LeRobot / ACT), semantic segmentation, object detection

Programming: C/C++, Python, Bash, MATLAB/Simulink

Tools: Git, Docker, Linux (Ubuntu), Fusion 360

Languages: French (native), English (C2)

Interests: INSA Lyon Robotics Club (Clubelek), OpenClaw, 3D printing (Bambu Lab P1S)